
SAE Roomba

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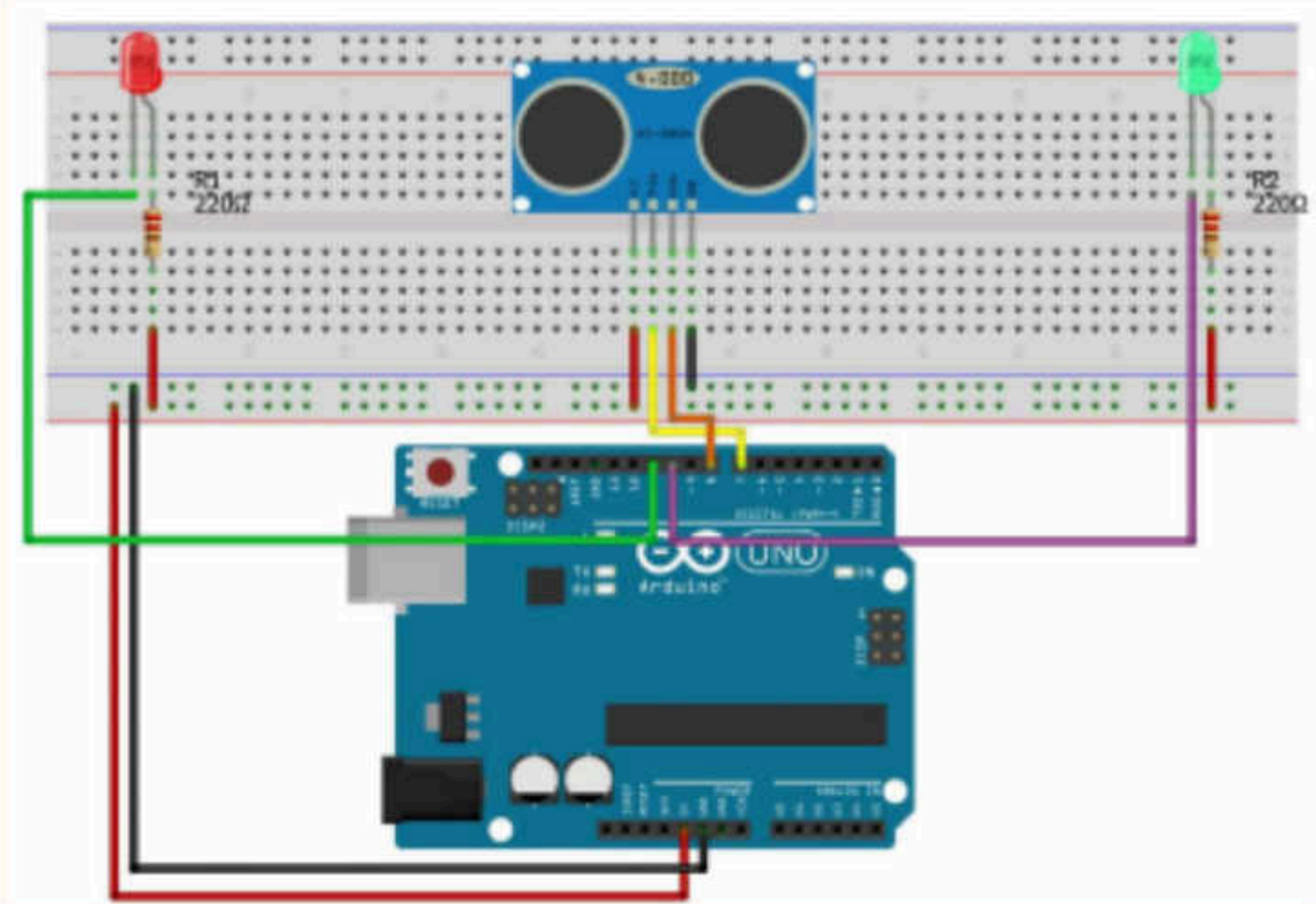
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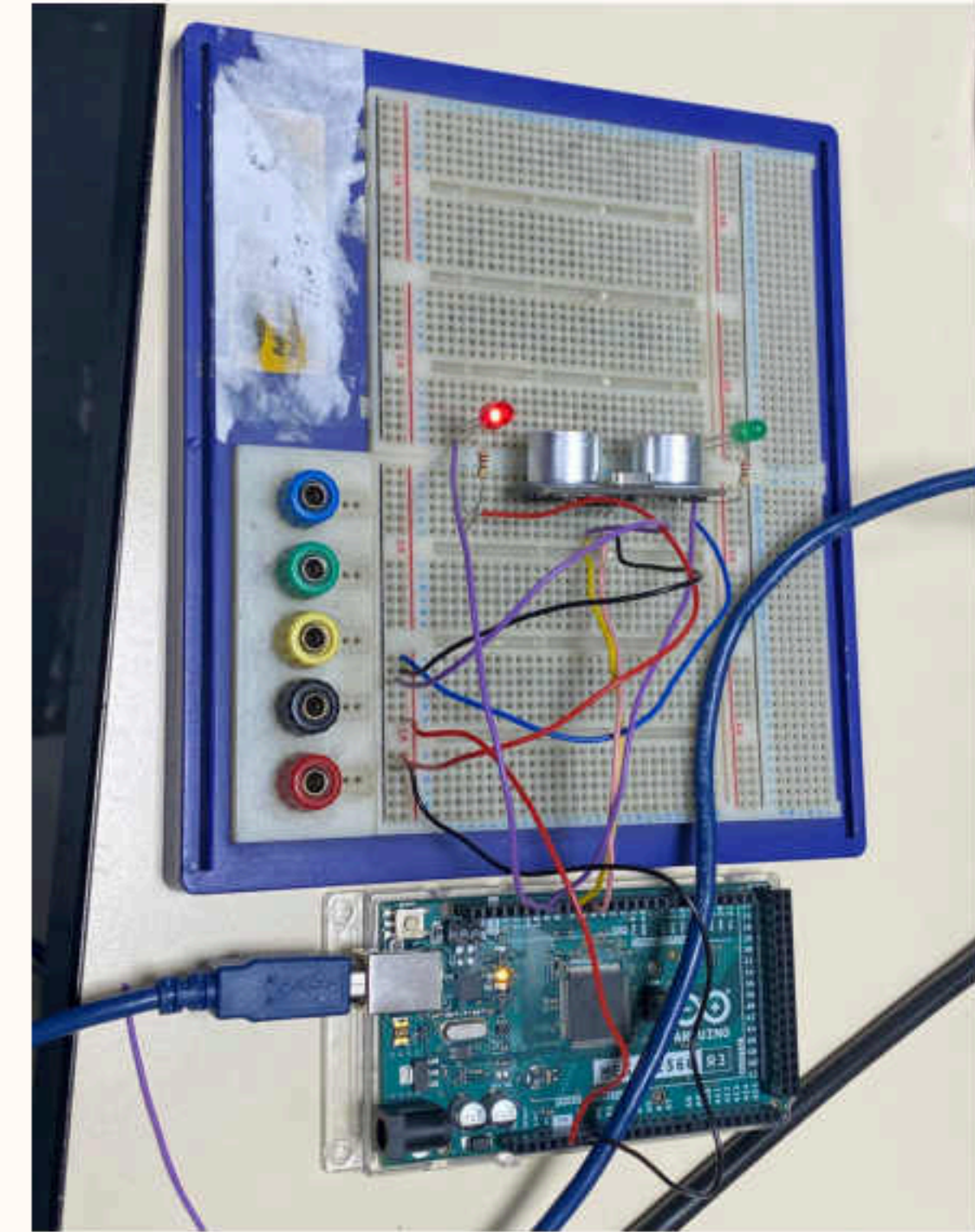
Introduction

Study of the vacuum detection block of a Roomba robot
Getting started with Arduino (ultrasonic sensor)
Validation (tests, measure, oscilloscope)

Carte Arduino Uno

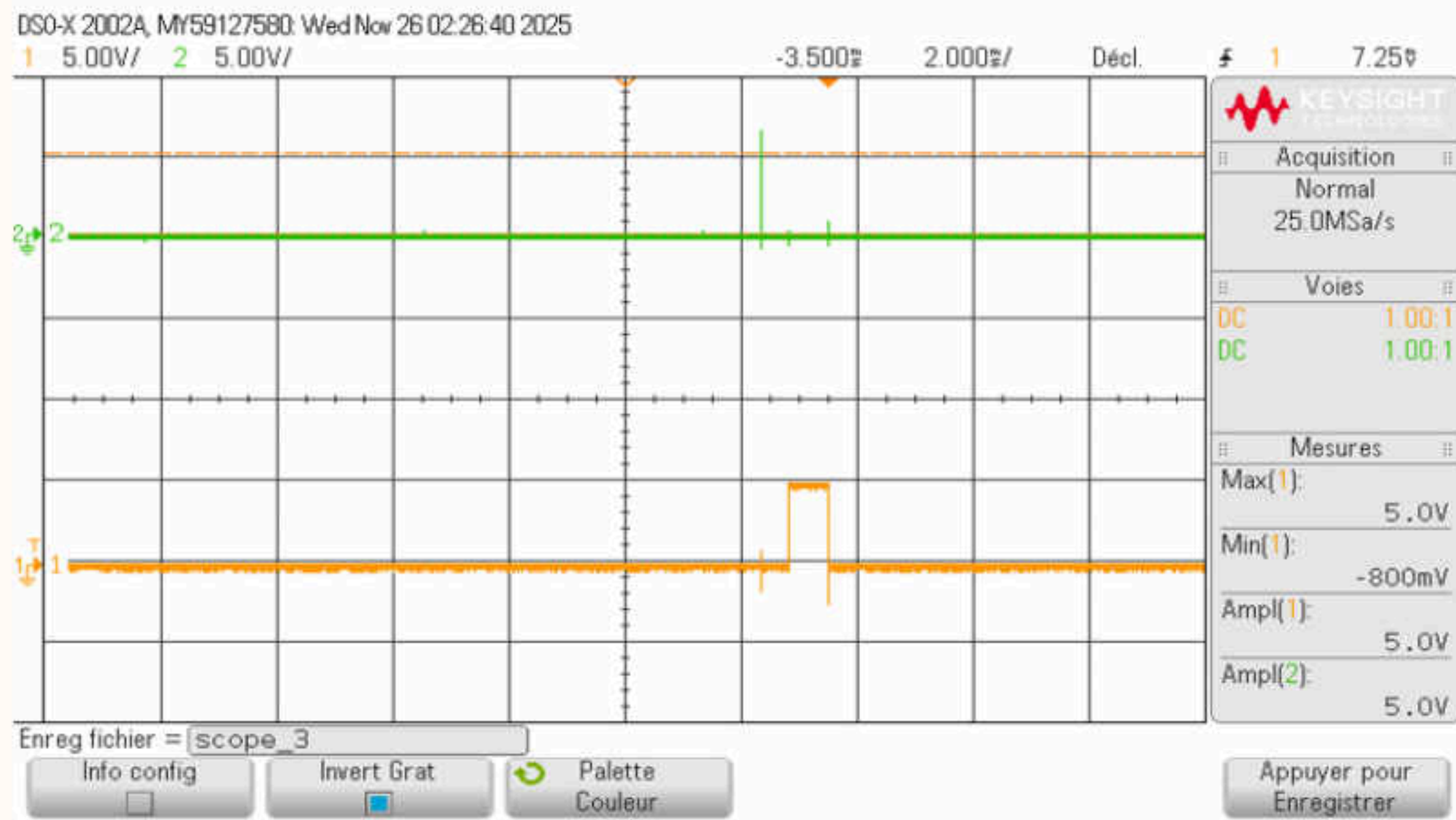


Tinkercard



Branchement

Carte Arduino Uno

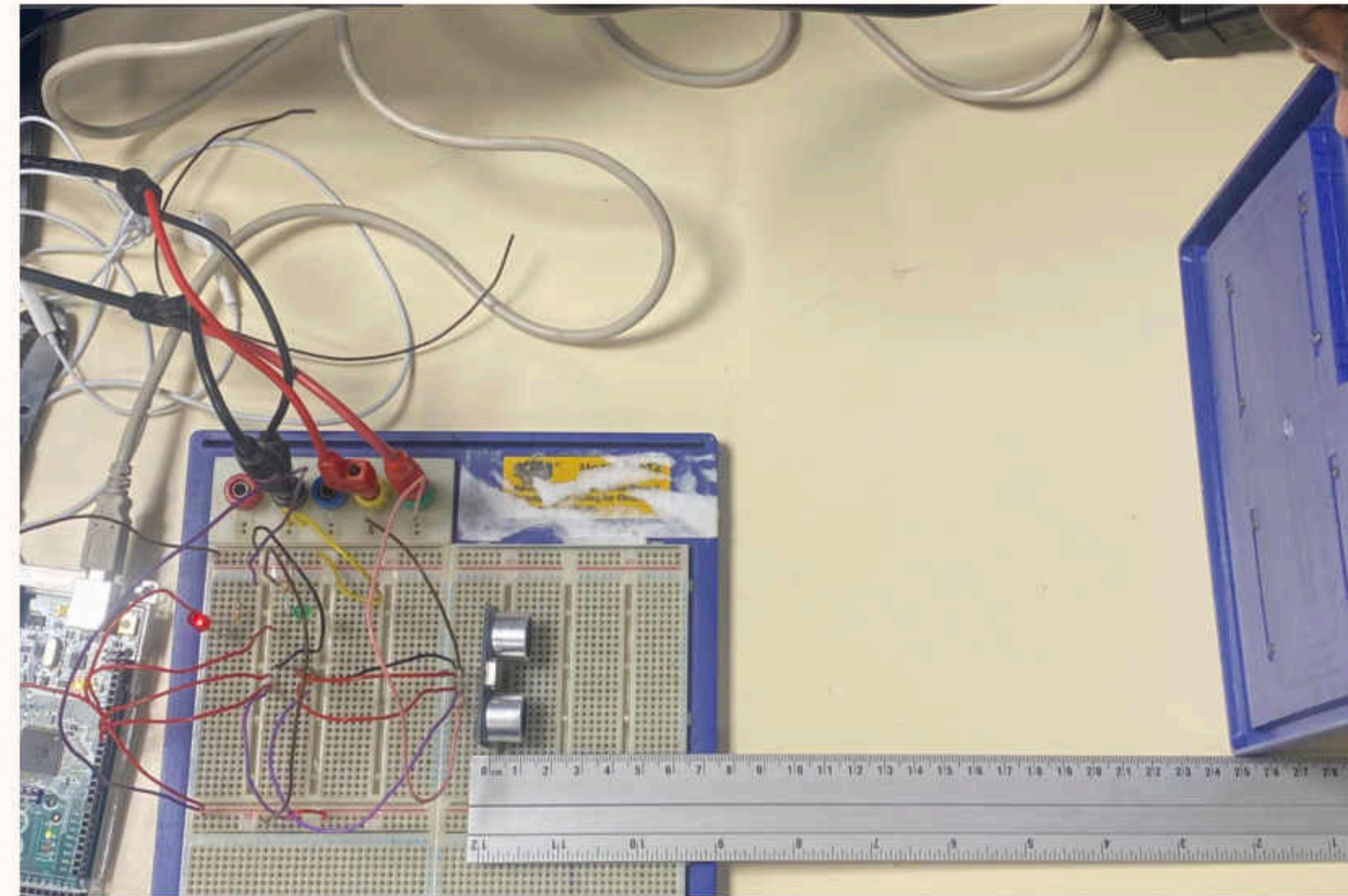
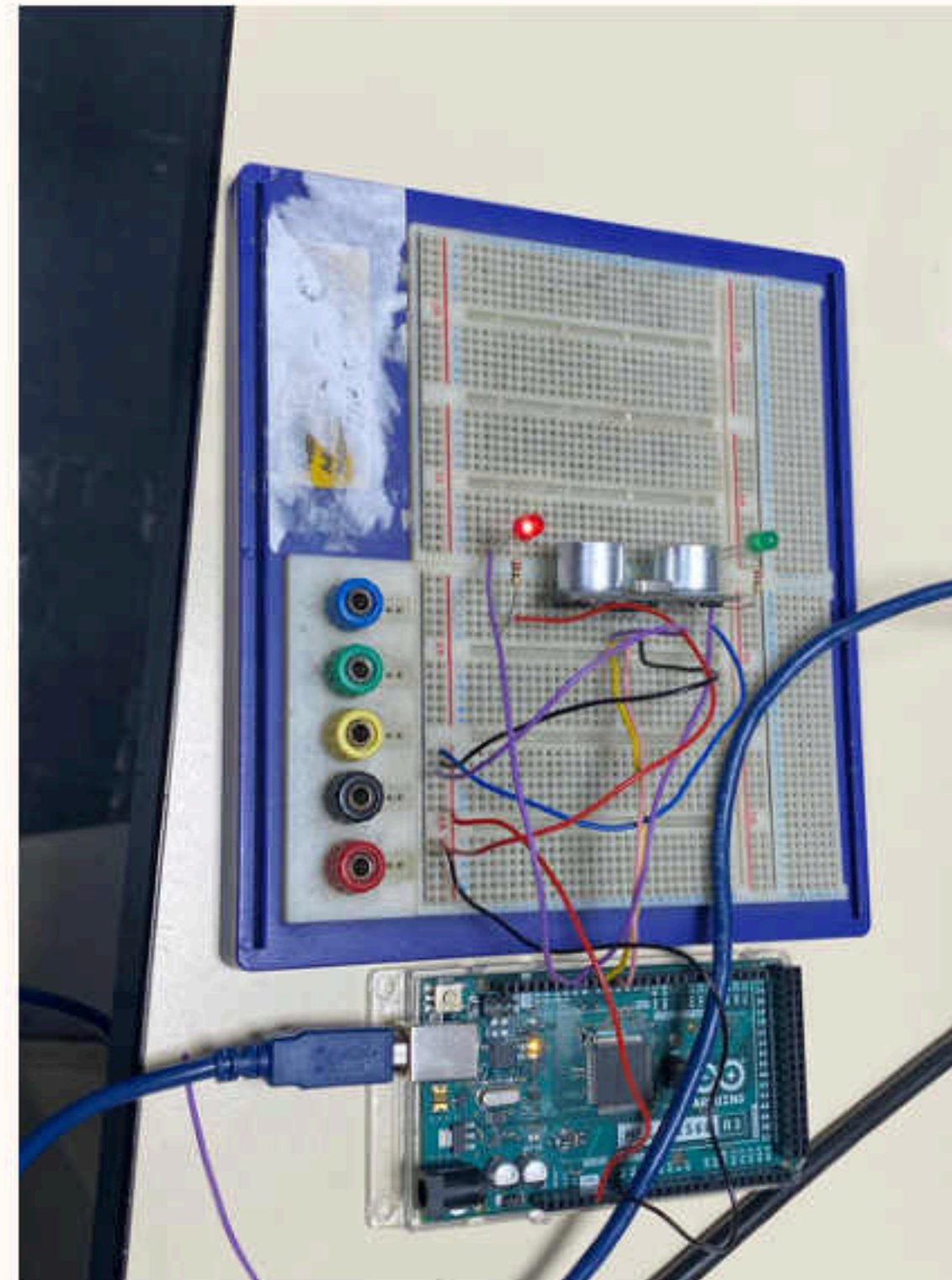
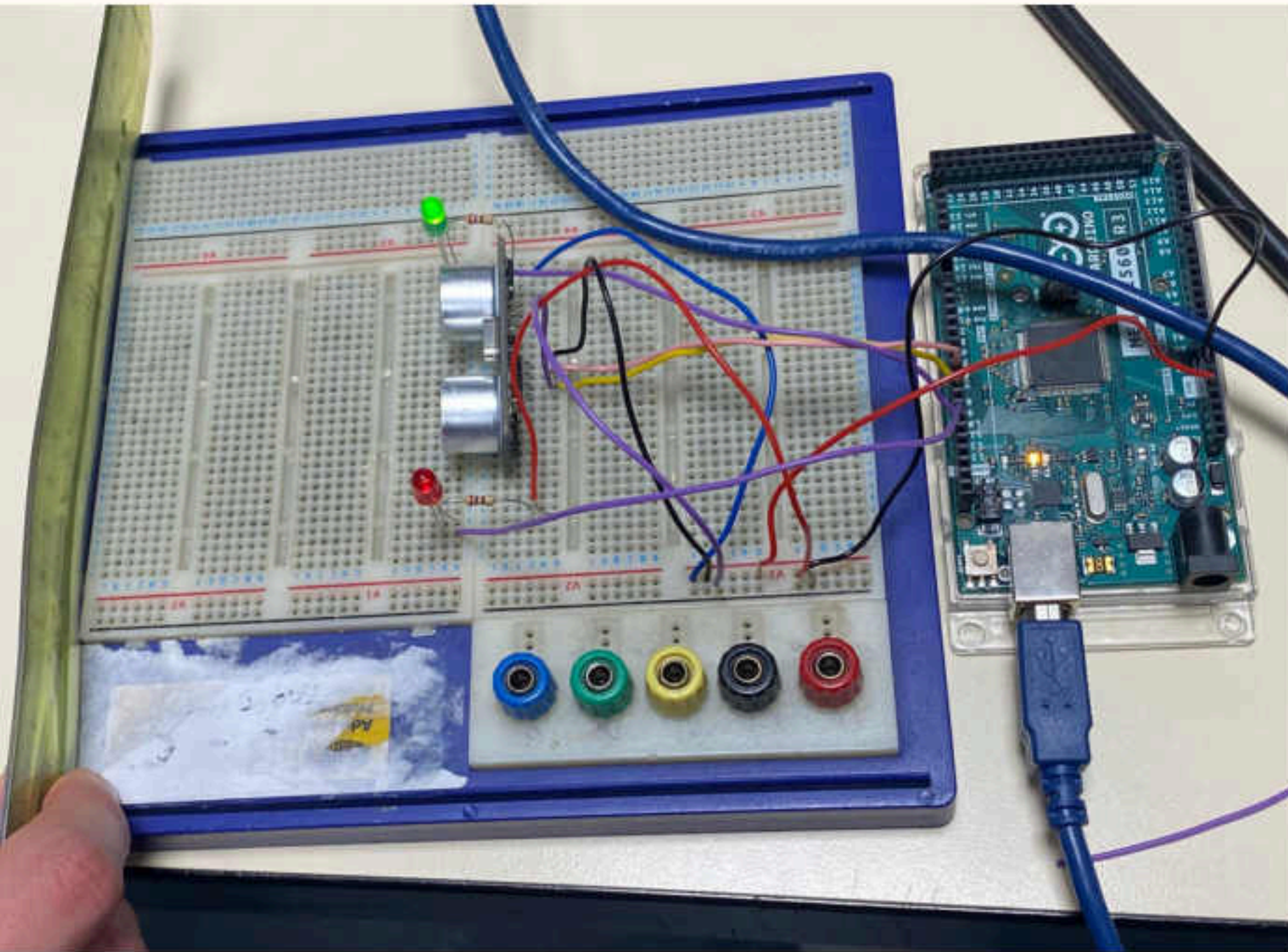


Obstacle proche



Obstacle éloigné

Test programme



Etude FS 3.1

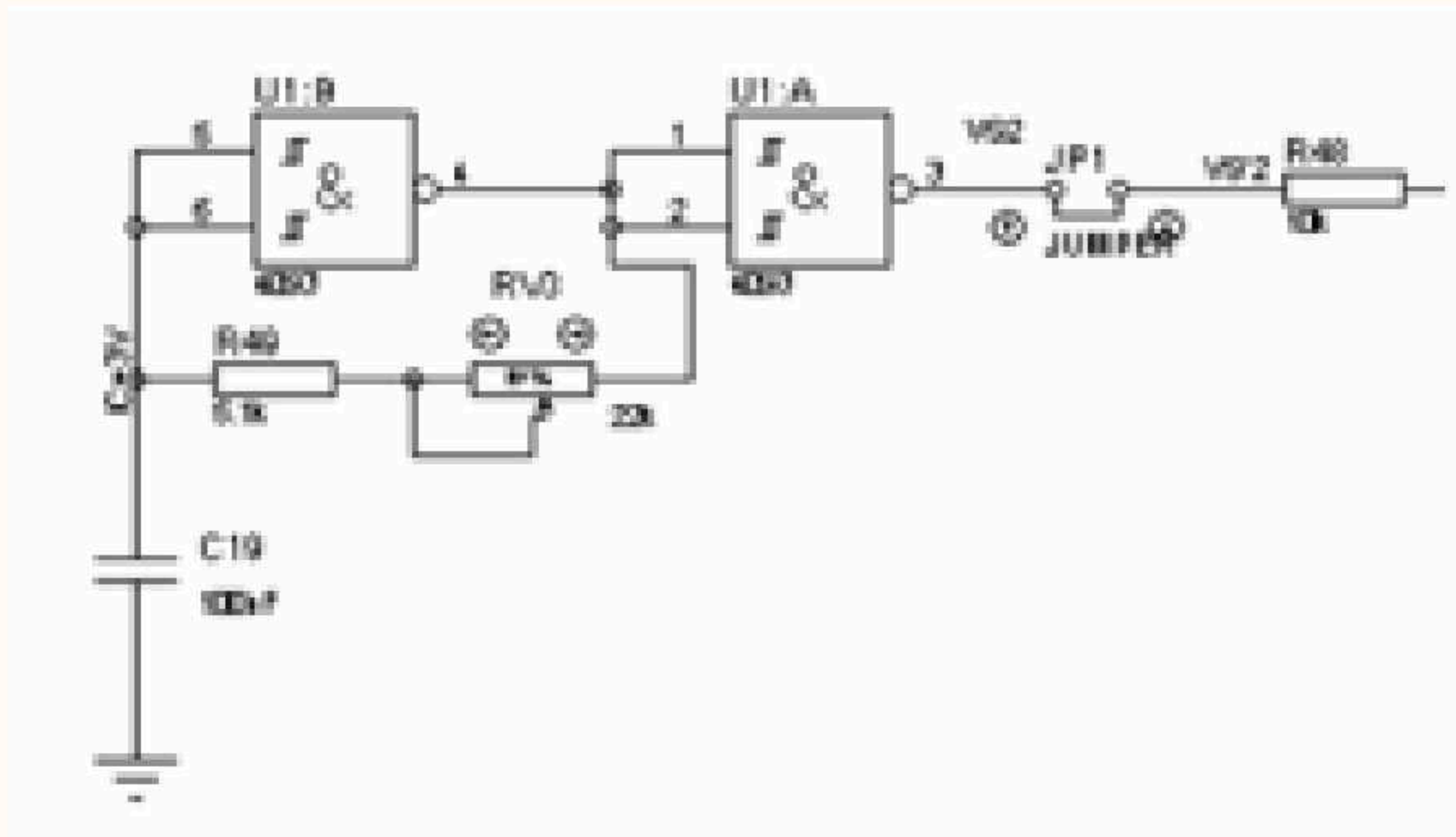
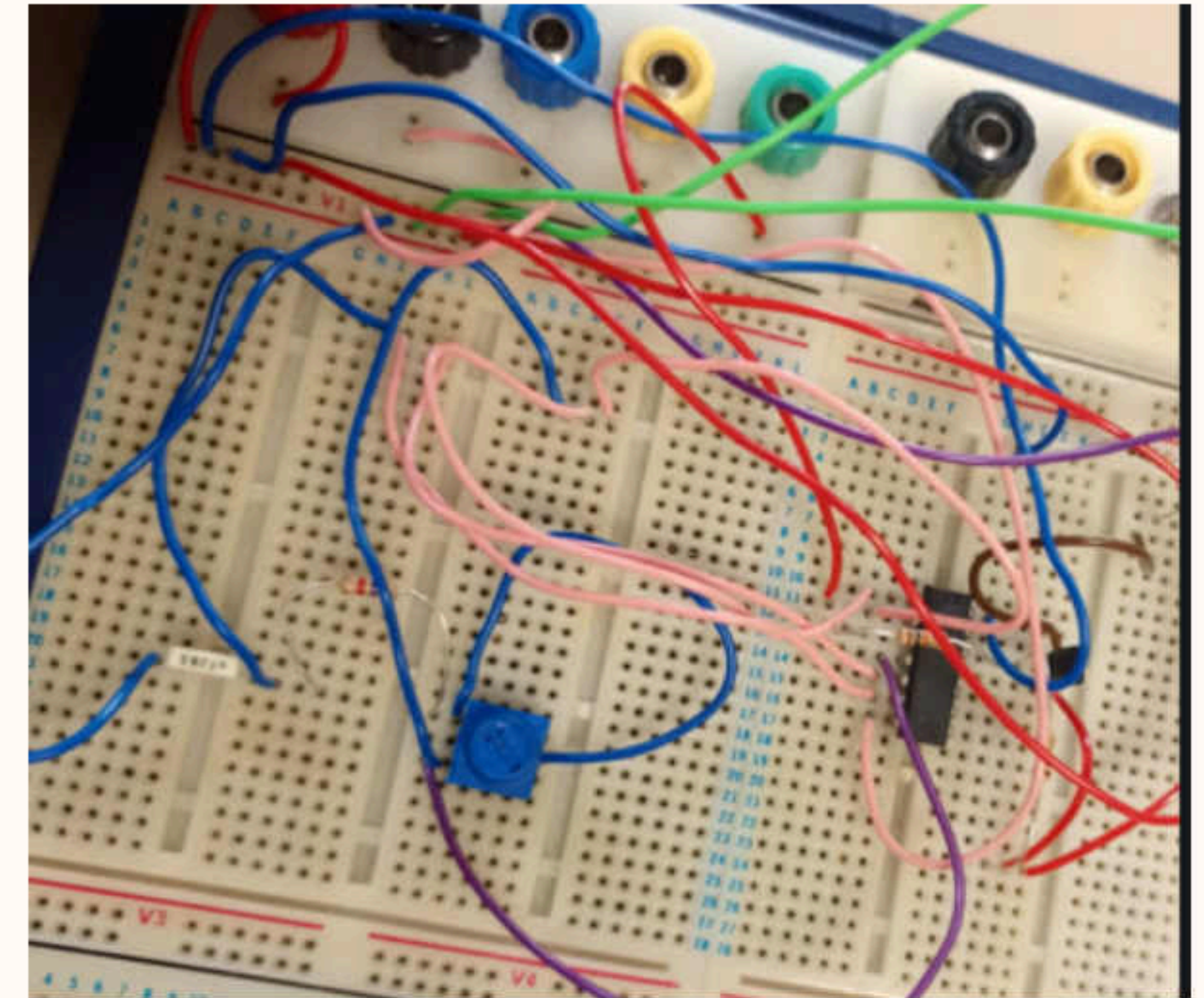
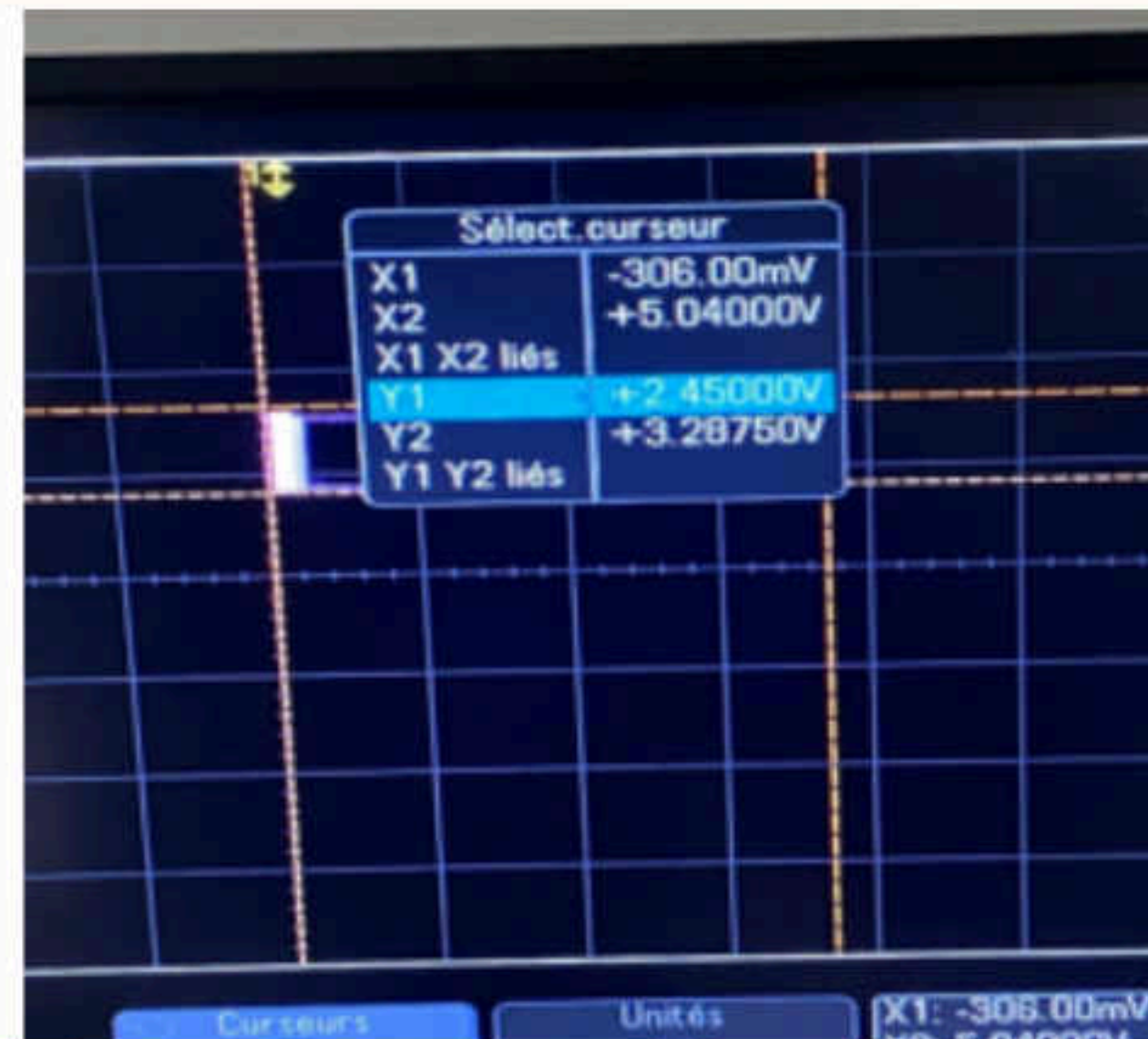
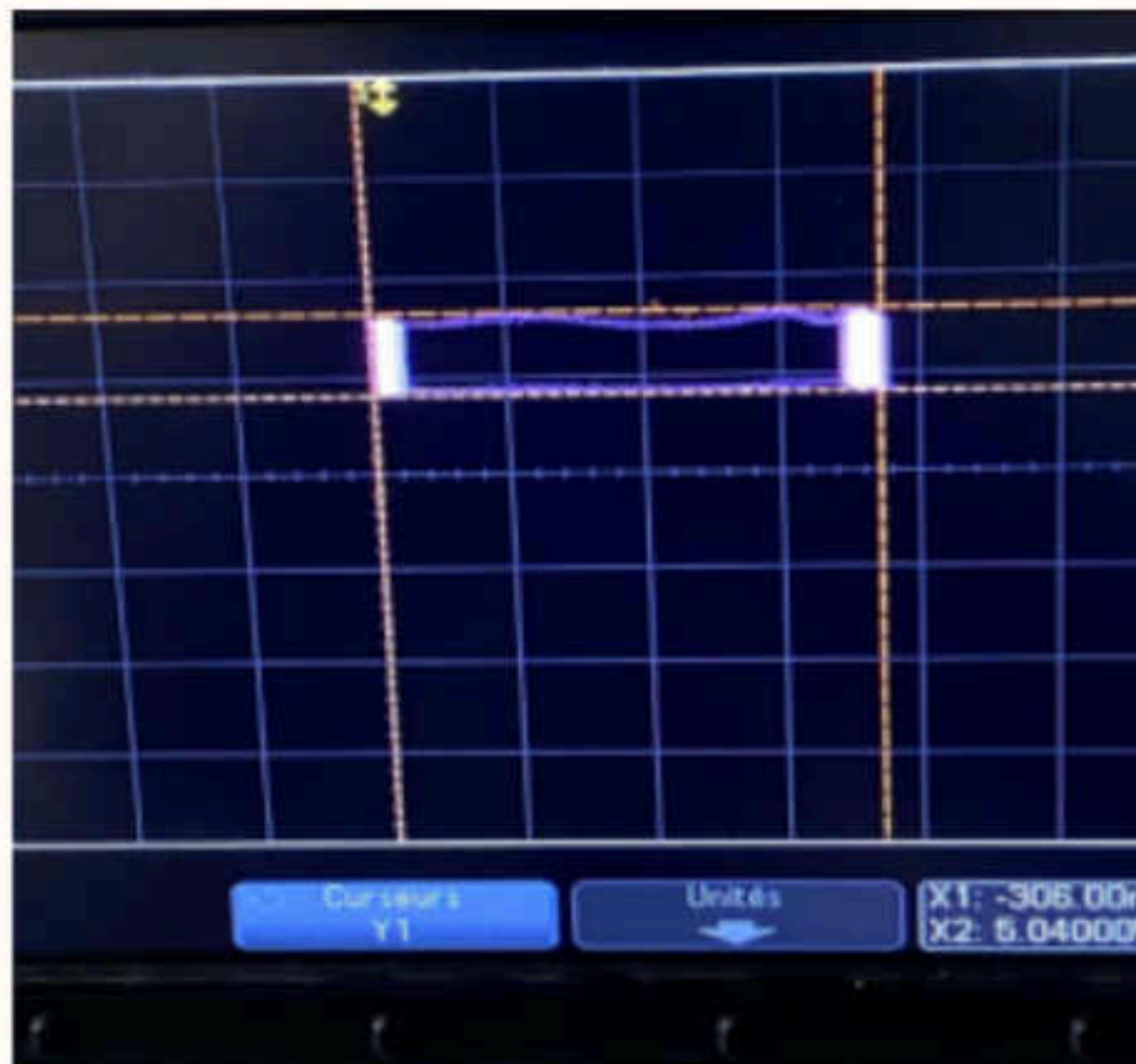
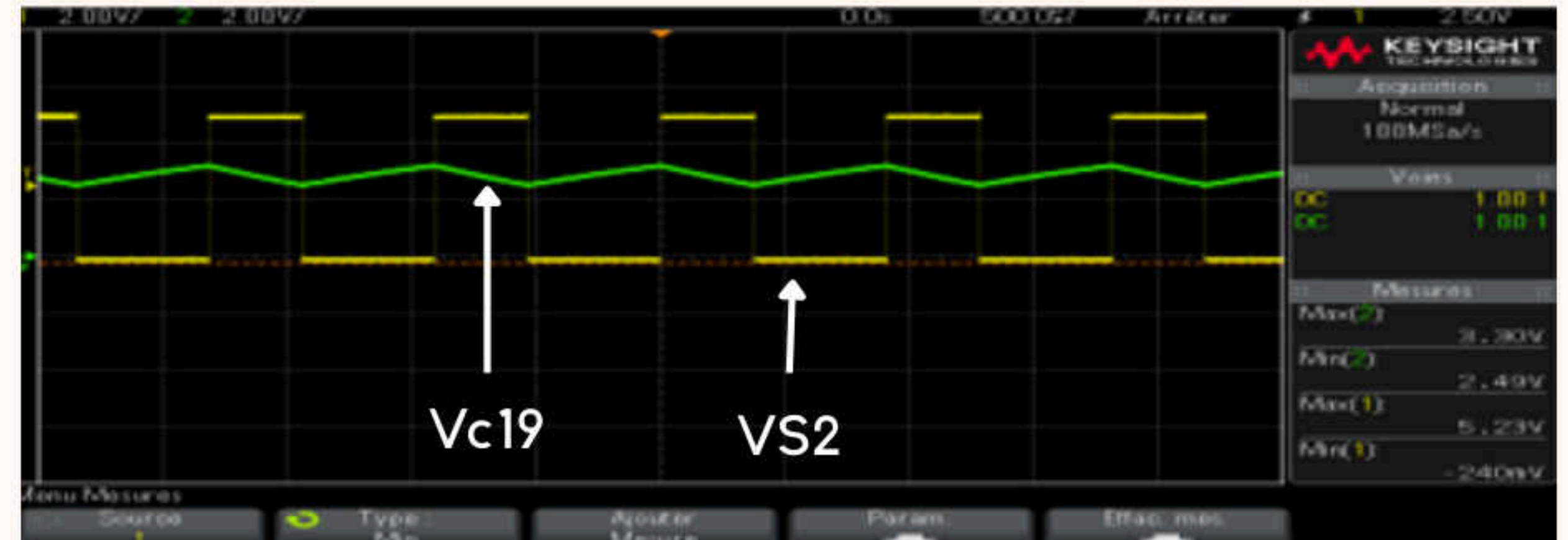


Schéma FS 3.1 (génération
signal modulant)



Branchement

Signaux FS 3.1



$$V_{\max} = V_c(t) = 3,28V$$

$$V_{\min} = V_{\text{init}} = 2,45V$$

$$RC = 16,8k\Omega$$

Formule Helmholtz :

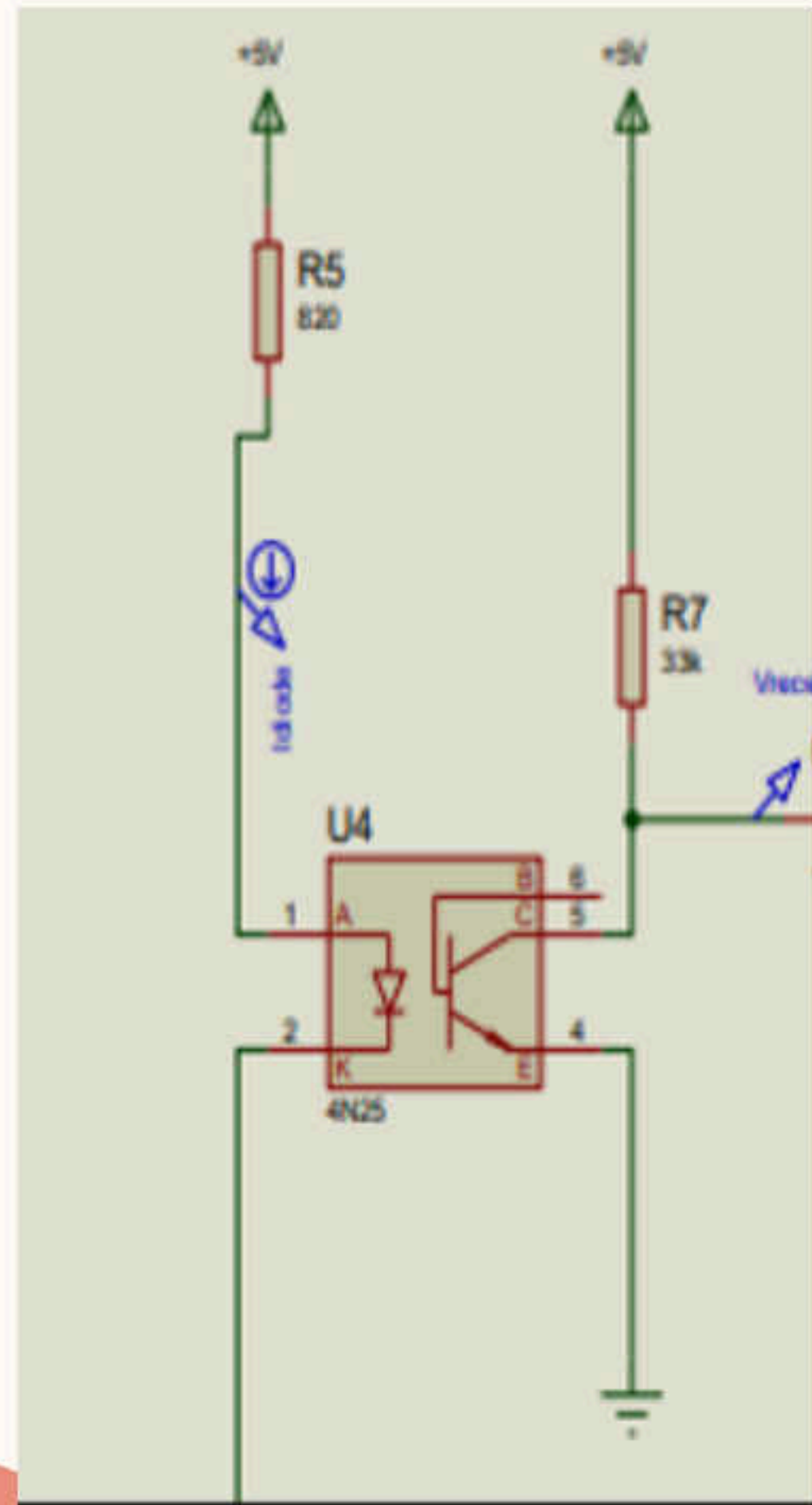
$$V_c(t) = V_{\text{final}} + (V_{\text{init}} - V_{\text{final}}) \times \exp.t/RC$$

$$V_c(t) = 3,22V$$

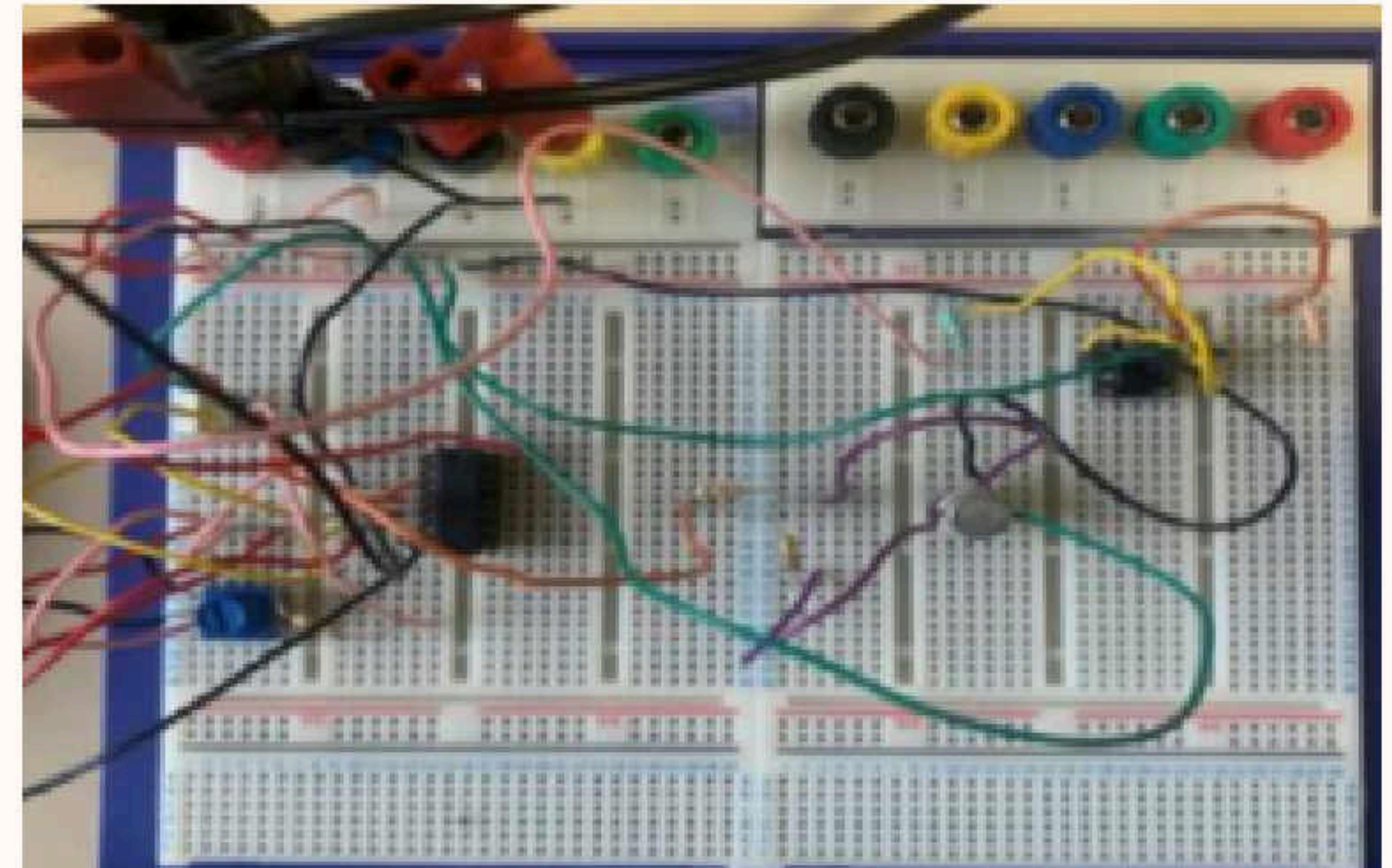
$$V_c(t) = V_{\max}$$

Etude FS 3.2 et FS 3.3

(émission infrarouge et réception infrarouge)



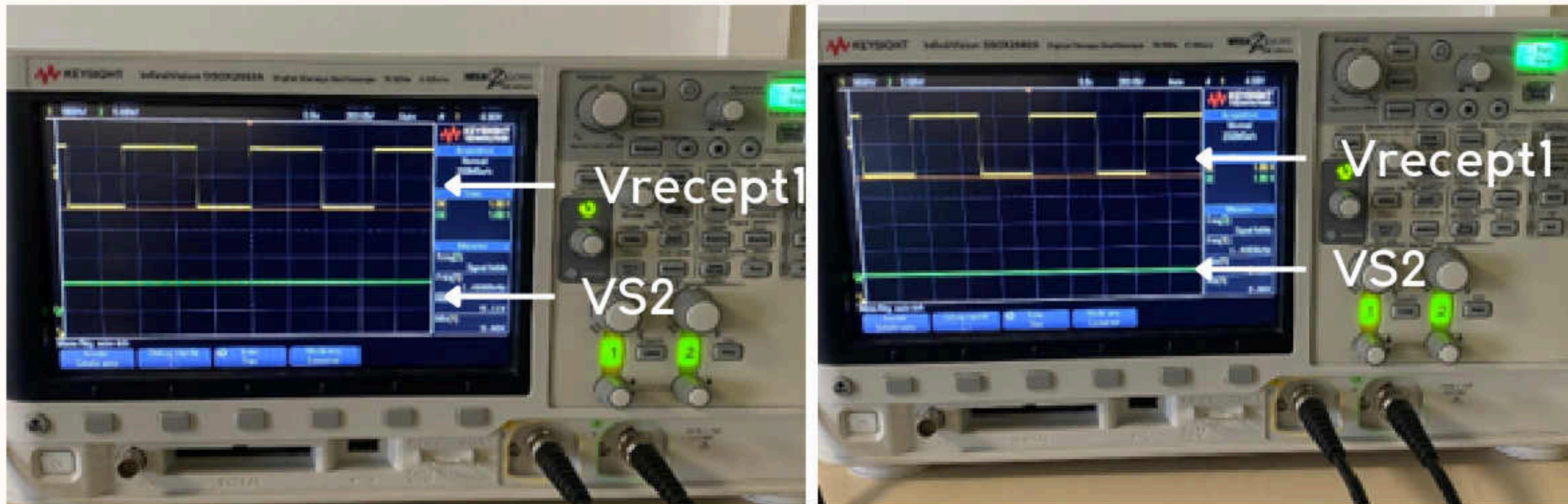
Schéma



Branchement

Etude FS 3.2 et FS 3.3

(émission infrarouge et réception infrarouge)



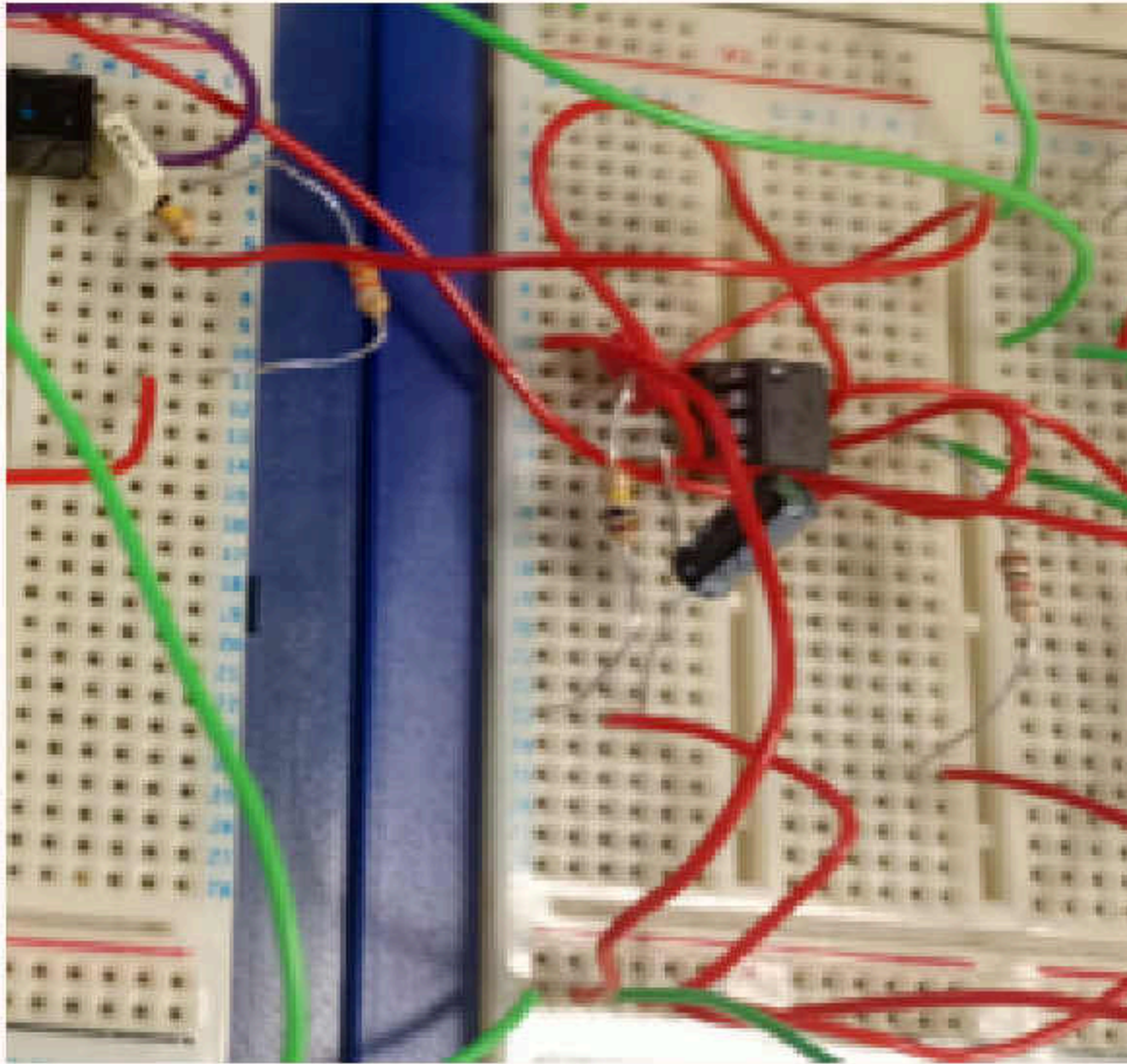
Signaux

Résumé du comportement de Vrecept1 :

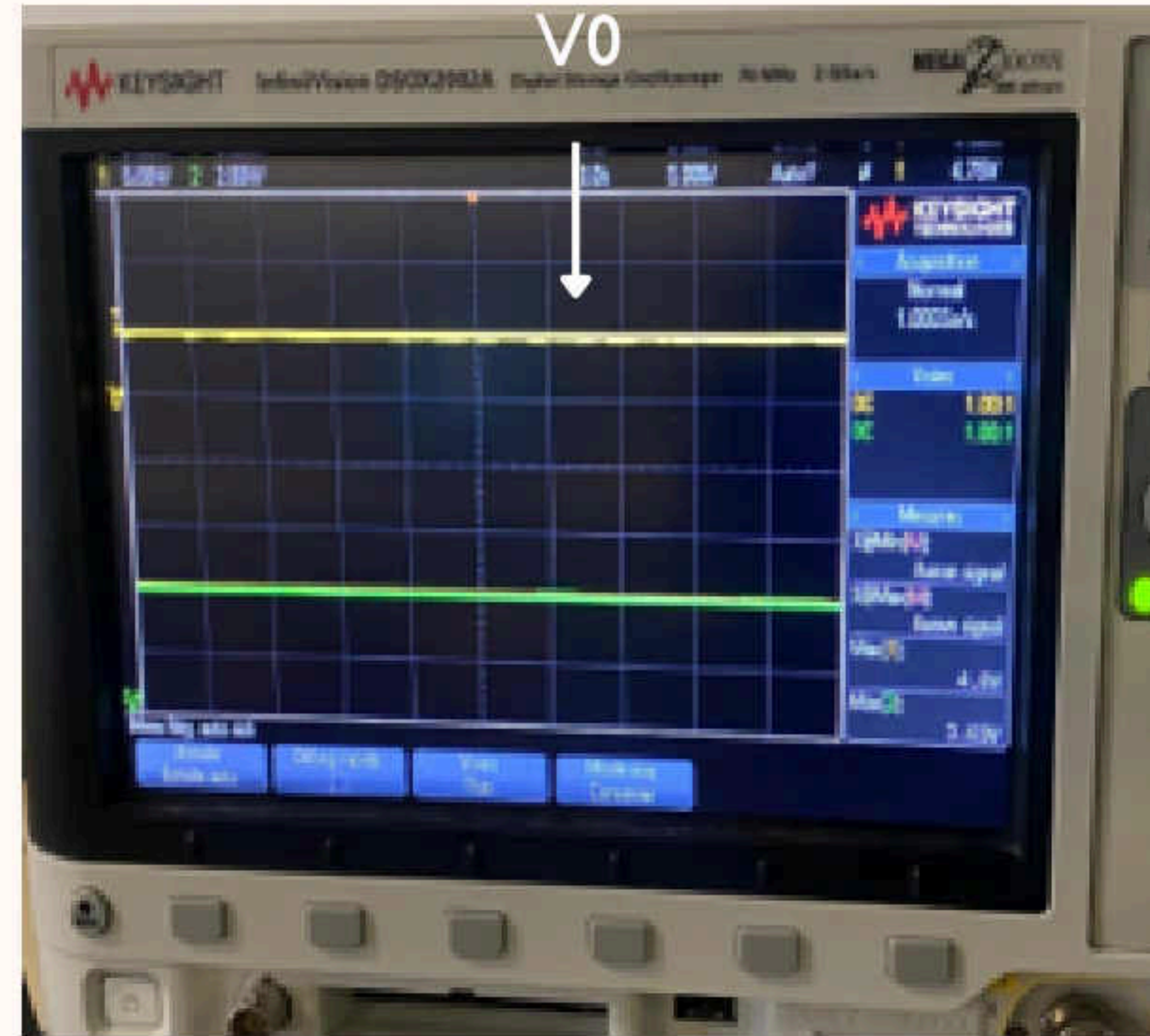
État de la LED IR	Présence d'une surface réfléchissante	État du phototransistor	Tension Vrecept1
Activée	Présente	Saturé	Faible (proche de 0V)
Activée	Absente (vide)	Bloqué	Élevée (proche de 5V)
Désactivée	Peu importe	Bloqué	Élevée (proche de 5V)

Etude FS 3.4

(conversion fréquence/tension)



Branchement

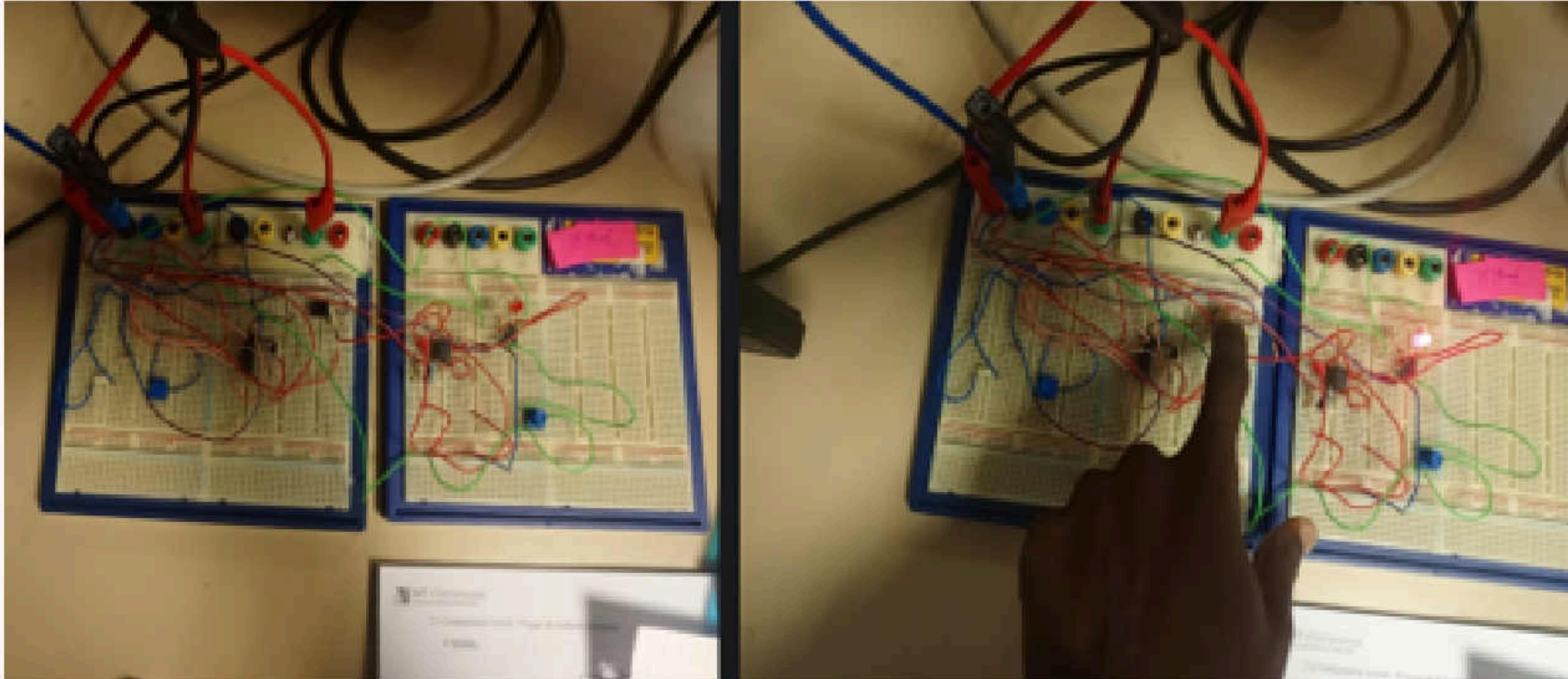


Signaux

$$V0 = V_{cc} * F(in) * C1 * R1$$
$$V0 = 4,1V$$

Etude FS 3.5

(comparaison)



Branchement

Conclusion

